RUNGTA INTERNATIONAL SKILLS UNIVERSITY

School of Information Technology • Bachelor of Computer Application (BCA — 1st Sem, Section 1A)
Programming Fundamentals Using Python (RU-100-05-00058) • Project Report: PS No. 4

Project Title	**PS No. 4: Quiz Application**	**Student Name**	Keerti Shukla
Course & Section	BCA 1st Semester / Section 1A	**Course Code**	RU-100-05-00058
Tools & Libraries	Python 3, `random` module	**Core Concepts**	Lists, Dictionaries, Loops, Random

1. Introduction & Project Requirements

This project was developed for the first-semester BCA Python programming evaluation (**PS No. 4: Quiz Application**). The objective is to create a console-based multiple-choice quiz application that stores at least 10 questions in a list of dictionaries, selects questions using Python's `random` module, validates every user answer (accepting only `A`, `B`, `C`, or `D`), keeps track of the score, and displays the final score, percentage, and `Pass`/`Fail` status based on a 50% passing mark.

Feature	Requirement in Question	How It Is Implemented in Python
Question Bank	Keep at least 10 questions in a list of dictionaries	Created `questions` list containing 10 dictionaries with keys `"question"`, `"options"`, and `"answer"`.
Random Selection	Select questions randomly using the `random` module	Used `random.sample(questions, total)` for random play and `questions[:3]` when `TEST_MODE = True`.
Input Validation	Accept only valid option letters (A, B, C, or D)	Used a `while True` loop checking `if answer in ["A", "B", "C", "D"]`; repeats the prompt on invalid input without losing an attempt.
Score Tracking	Check every answer and keep the score	Compared `answer == q["answer"]`; incremented `score += 1` on correct answers and showed the right answer if wrong.
Result & Pass/Fail	Show final score, percentage, and pass/fail message	Calculated `percentage = (score / total) * 100`; printed `Pass` if `percentage >= 50`, else `Fail`.

2. Logic Flow & Score Analysis

1. **Step 1 (Setup Question Bank):** Store 10 multiple-choice questions in a Python list of dictionaries. Each dictionary stores the question text, a nested dictionary of 4 options (`A`, `B`, `C`, `D`), and the correct option key.
2. **Step 2 (Select Questions):** Initialize `score = 0` and `total = 3`. If `TEST_MODE` is enabled, select the first 3 syllabus test questions in order; otherwise call `random.sample(questions, total)`.
3. **Step 3 (Display & Validate Input):** Loop through the selected questions using `enumerate()`. Print the question and options, then use a `while True` loop with `.strip().upper()` to ensure the user enters only `A`, `B`, `C`, or `D`.
4. **Step 4 (Check Answer):** If the user's input matches `q["answer"]`, print `Correct!` and add 1 to `score`. Otherwise, print the correct answer for that question.
5. **Step 5 (Calculate Final Result):** Compute `percentage = (score / total) * 100`. Display the score, percentage formatted to 2 decimal places (`:.2f`), and `Result: Pass` (if ≥ 50%) or `Result: Fail`.

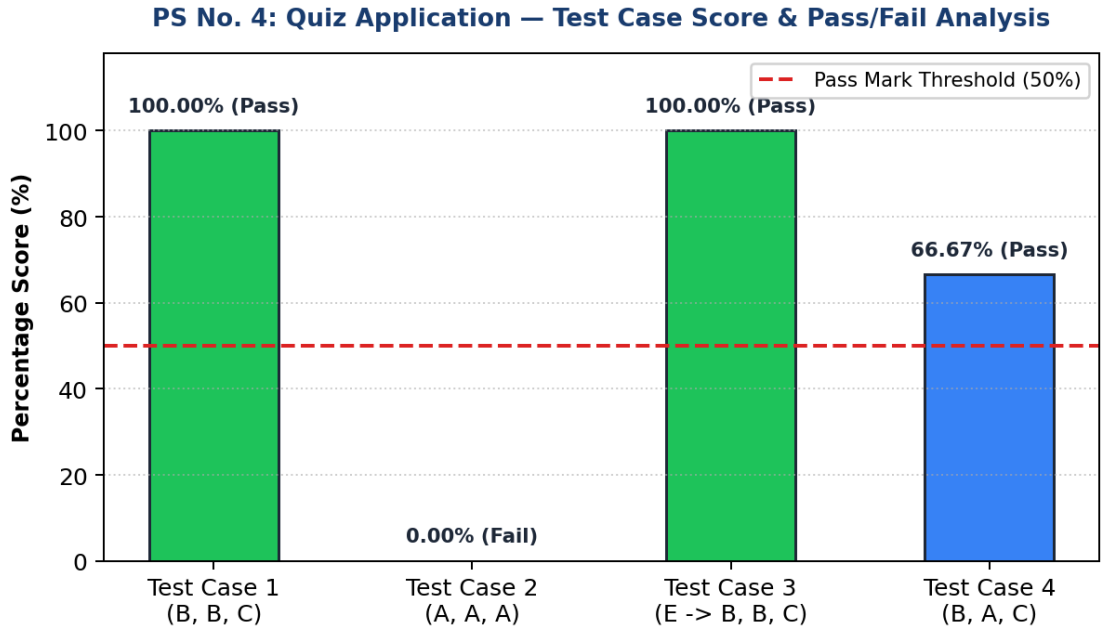


Figure 1: Percentage scores across the 4 official evaluation test cases compared against the 50% pass threshold.

3. Official Test Cases & Verification

In test mode, the 3 questions asked are: Q1 *'Keyword to define a function?'* (Ans: `B`), Q2 *'len("Hi") = ?'* (Ans: `B`), and Q3 *'Mutable type?'* (Ans: `C`). Below are the results for all 4 official test cases:

Test Case	Input	Expected Output	Program Output & Status
Test Case 1 (Normal Input)	`Answers: B, B, C`	`Score: 3/3 | Percentage: 100.00% Result: Pass`	`Score: 3/3 | Percentage: 100.00% Result: Pass` **Status: PASS**

Test Case	Input	Expected Output	Program Output & Status
Test Case 2 (Boundary Input)	`Answers: A, A, A`	`Score: 0/3 | Percentage: 0.00% Result: Fail Correct answers: B, B, C`	`Score: 0/3 | Percentage: 0.00% Result: Fail Correct answers: B, B, C` **Status: PASS**
Test Case 3 (Invalid Input)	`Q1 answer: E, then B`	`Invalid option. Enter A, B, C or D. (Q1 is asked again) Correct!`	`Invalid option. Enter A, B, C or D. Correct!` **Status: PASS**
Test Case 4 (Different Valid)	`Answers: B, A, C`	`Q2 Wrong. Correct answer: B Score: 2/3 | Percentage: 66.67% | Result: Pass`	`Q2 Wrong. Correct answer: B Score: 2/3 | Percentage: 66.67% Result: Pass` **Status: PASS**

4. Viva Preparation Questions & Answers

- **Q1: How did you store the questions and options in your program?**
  Ans: I used a list of dictionaries. Each dictionary represents one question with three keys: `"question"` (string), `"options"` (a nested dictionary mapping A, B, C, D to their text), and `"answer"` (the correct option letter).
- **Q2: Why did you use `random.sample()` instead of `random.choice()`?**
  Ans: `random.sample(questions, total)` picks unique questions without repetition, whereas `random.choice()` picks only one item and could repeat the same question if called in a loop.
- **Q3: How does your program handle invalid inputs like `E` or lowercase `b`?**
  Ans: We use `.strip().upper()` to convert lowercase letters like `b` into `B` automatically. If the user enters an invalid letter like `E`, the `while True` loop prints `Invalid option. Enter A, B, C or D.` and asks for the answer again without deducting any score.

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5. Complete Python Source Code (`quiz_application.py`)

```
# PS No. 4: Quiz Application
# Student Name: Keerti Shukla
# Course: BCA 1st Semester (Section 1A)
# Subject: Programming Fundamentals Using Python (RU-100-05-00058)
# University: Rungta International Skills University

import random

TEST_MODE = True

# Question Bank (10 questions stored as a list of dictionaries)
questions = [
```

```
    {
        "question": "Keyword to define a function?",
        "options": {"A": "func", "B": "def", "C": "function", "D": "lambda"},
        "answer": "B"
    },
    {
        "question": 'len("Hi") = ?',
        "options": {"A": "1", "B": "2", "C": "3", "D": "4"},
        "answer": "B"
    },
    {
        "question": "Mutable type?",
        "options": {"A": "tuple", "B": "str", "C": "list", "D": "int"},
        "answer": "C"
    },
    {
        "question": "Which symbol is used for comments?",
        "options": {"A": "//", "B": "#", "C": "/*", "D": "$"},
        "answer": "B"
    },
    {
        "question": "Which function displays output?",
        "options": {"A": "input()", "B": "print()", "C": "show()", "D": "display()"},
        "answer": "B"
    },
    {
        "question": "Which loop is used to iterate over a sequence?",
        "options": {"A": "if", "B": "for", "C": "def", "D": "try"},
        "answer": "B"
    },
    {
        "question": "Which data type stores True or False?",
        "options": {"A": "int", "B": "str", "C": "bool", "D": "list"},
        "answer": "C"
    },
    {
        "question": "Which operator is used for exponentiation?",
        "options": {"A": "%", "B": "//", "C": "**", "D": "/"},
        "answer": "C"
    },
    {
        "question": "Which keyword is used to import a module?",
        "options": {"A": "include", "B": "import", "C": "module", "D": "load"},
        "answer": "B"
    },
    {
        "question": "Which function returns the largest value?",
        "options": {"A": "min()", "B": "sum()", "C": "max()", "D": "len()"},
        "answer": "C"
    }
]


def run_quiz():
    score = 0
    total = 3

    if TEST_MODE:
        selected_questions = questions[:3]
    else:
        selected_questions = random.sample(questions, total)

    print("\n===== WELCOME TO QUIZ APPLICATION =====")
    print("Total Questions:", total)
    print("Pass Marks: 50%")
    print("----------------------------------------")

    for i, q in enumerate(selected_questions):
```

```
        print(f"\nQ{i + 1}: {q['question']}")

        for key, value in q["options"].items():
            print(f"{key}. {value}")

        while True:
            answer = input("Enter your answer (A/B/C/D): ").strip().upper()

            if answer in ["A", "B", "C", "D"]:
                break
            else:
                print("Invalid option. Enter A, B, C or D.")

        if answer == q["answer"]:
            print("Correct!")
            score += 1
        else:
            print(f"Q{i + 1} Wrong. Correct answer: {q['answer']}")

    percentage = (score / total) * 100

    print("\n========== QUIZ RESULT ==========")
    print(f"Score: {score}/{total} | Percentage: {percentage:.2f}%")

    if percentage >= 50:
        print("Result: Pass")
    else:
        print("Result: Fail")
        correct_list = ", ".join([q["answer"] for q in selected_questions])
        print(f"Correct answers: {correct_list}")

    print("================================")


if __name__ == "__main__":
    run_quiz()
```